Splunk SIEM Lab & Threat Detection Analysis

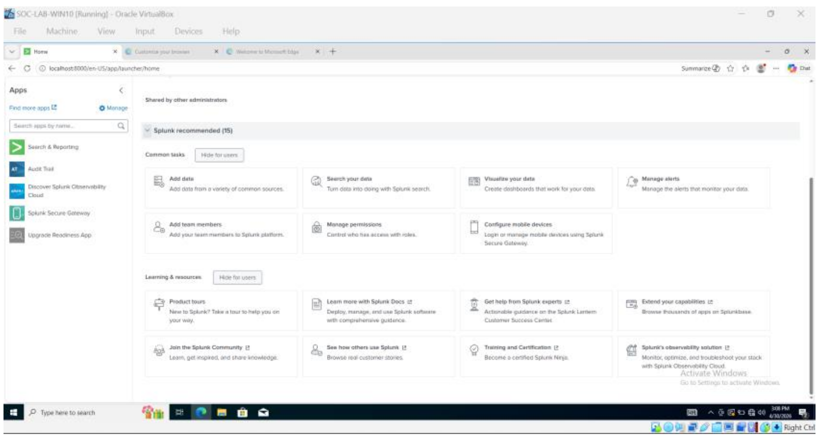
Project Overview

This project demonstrates the setup and configuration of a Security Information and Event Management (SIEM) system using Splunk. The objective was to ingest, monitor, and analyze Windows and Sysmon logs to gain visibility into endpoint activity and support security detection capabilities.

Screenshots

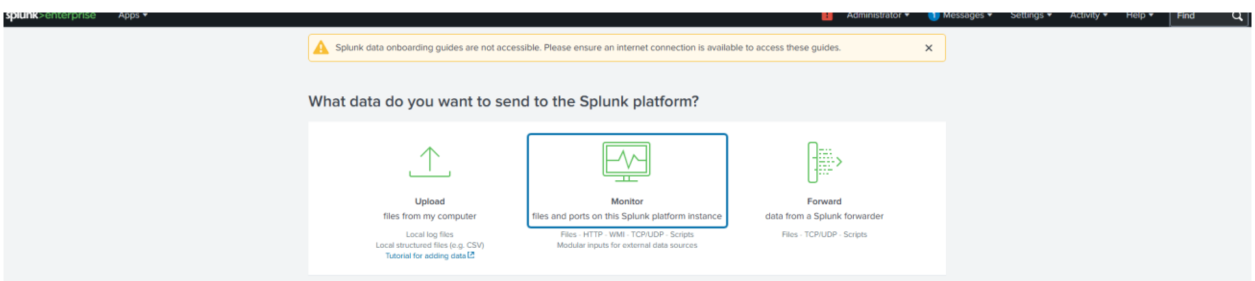
1. Splunk login or dashboard homepage

Purpose: Confirms successful deployment and accessibility of the SIEM platform



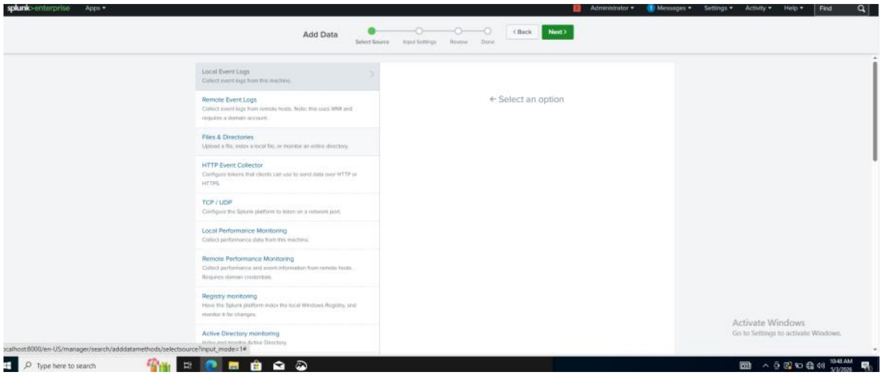
2. Selecting data input method (Monitor)

Purpose: Shows how data ingestion is initiated in Splunk



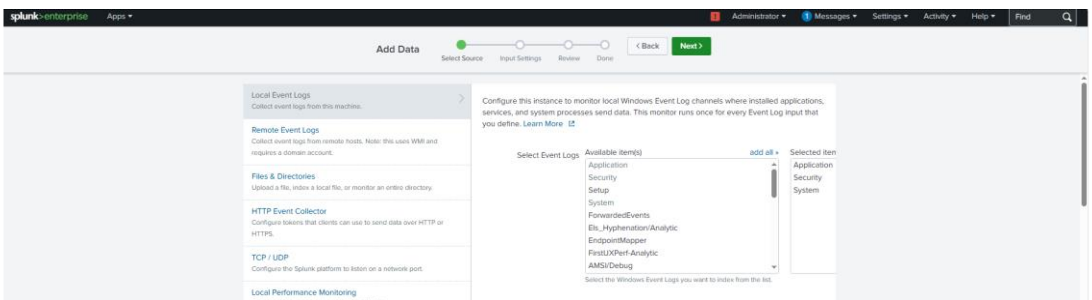
3. Selecting local Windows event logs for ingestion

Purpose: Demonstrates selection of Windows log source



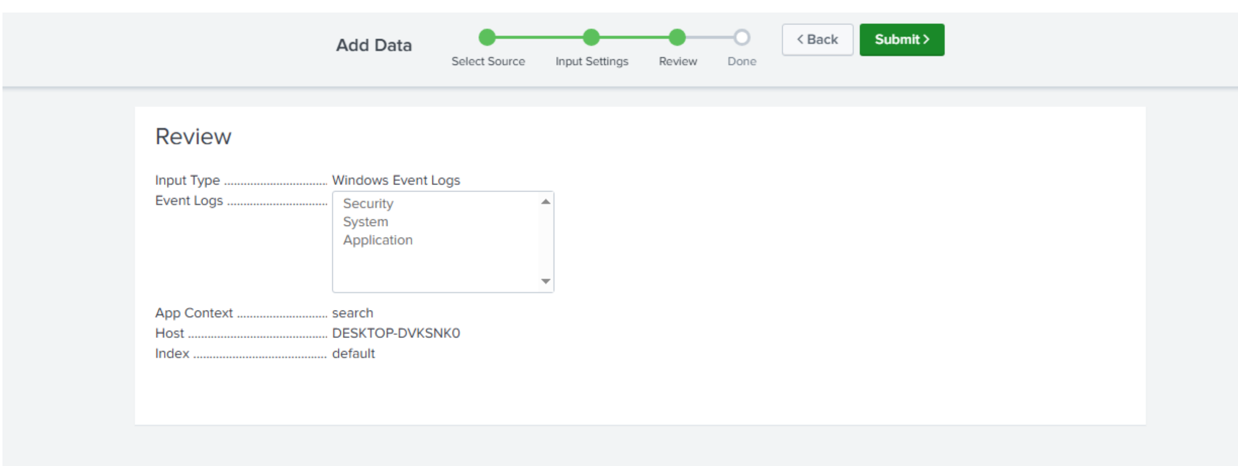
4. Configured Windows Event Logs for ingestion

Purpose: Shows configuration of log sources for monitoring



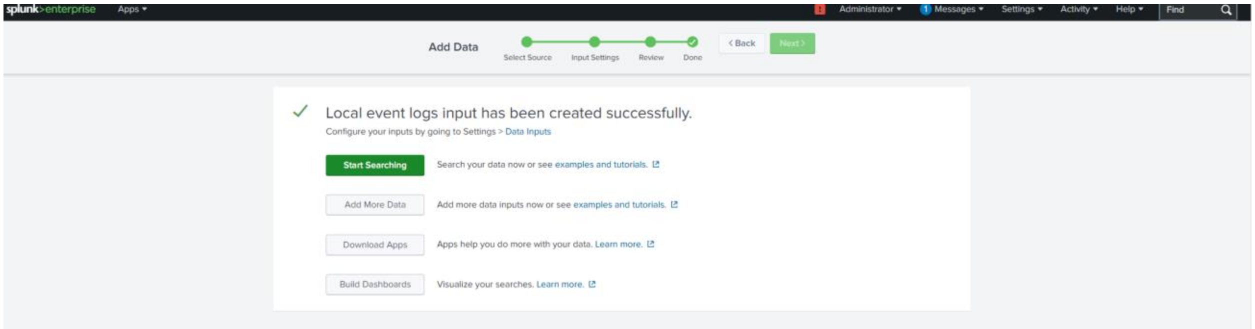
5. Reviewing configured Windows Event Logs before submission (5A)

Purpose: Shows validation of correct configuration prior to ingestion



5. Successfully configured Windows Event Log ingestion in Splunk (5B)

Purpose: Confirms successful deployment of log ingestion pipeline

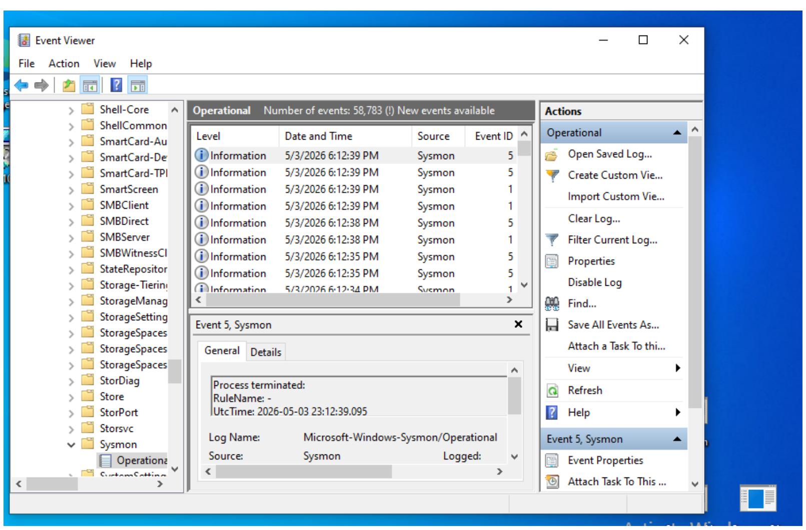


6A. Configured Sysmon log ingestion for advanced endpoint monitoring

Purpose: Demonstrates collection of advanced endpoint telemetry including process creation, command-line activity, and system behavior using Sysmon

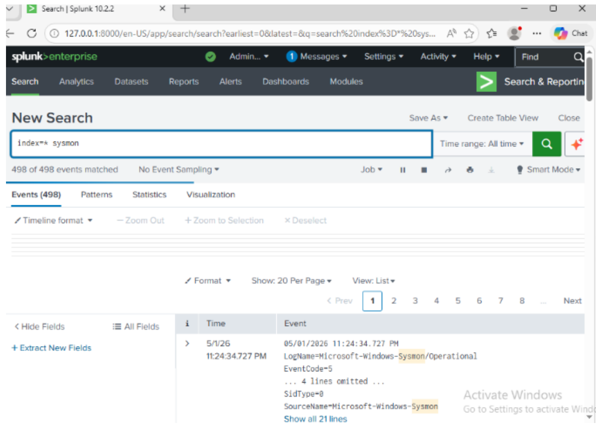
6B. Sysmon operational logs successfully generated in Event Viewer

Purpose: Confirms endpoint telemetry is active prior to SIEM ingestion



7. Sysmon logs successfully ingested and analyzed in Splunk

Purpose: Demonstrates real-time visibility into endpoint activity and the ability to detect process execution using SIEM analytics



Conclusion

Successfully configured a SIEM environment capable of ingesting and analyzing Windows and Sysmon logs. This project demonstrates foundational SOC analyst skills including log ingestion, endpoint monitoring, troubleshooting, and basic detection capability. This project simulates real-world SOC workflows including log ingestion, monitoring, troubleshooting, and basic detection analysis.

Troubleshooting & Problem Solving

- Resolved Splunk Web ingestion issues (404 errors) through troubleshooting and service validation
- Implemented manual log ingestion using inputs.conf when UI-based configuration failed
- Diagnosed and resolved Splunk service restart and permission-related issues
- Validated data ingestion pipeline using SPL queries (index=*, index=* sysmon)
- Simulated endpoint activity to confirm log generation and ingestion accuracy

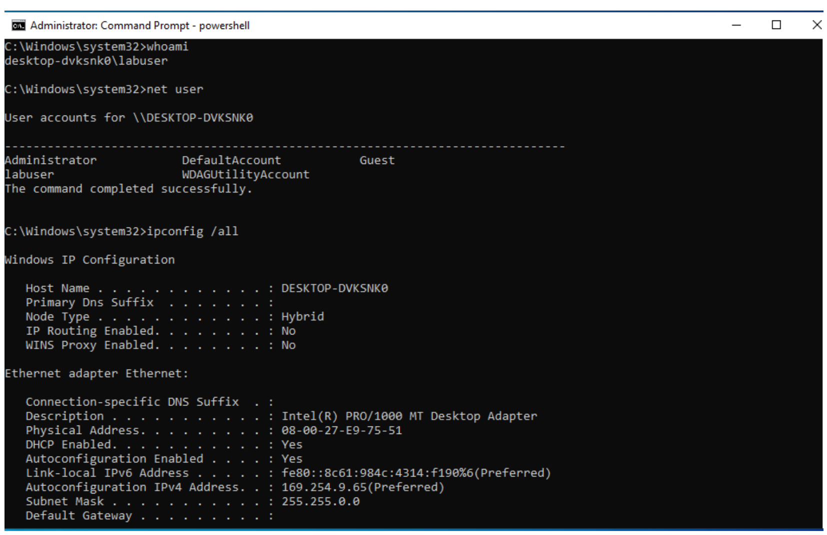
Threat Simulation & Detection Analysis

Simulated endpoint activity by executing various system and administrative commands (e.g., whoami, ipconfig, netstat, and PowerShell commands).

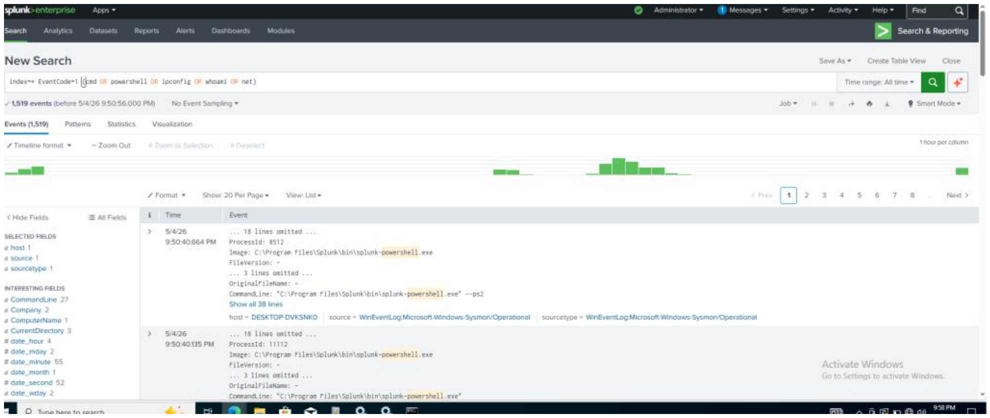
Leveraged Splunk SIEM to analyze Sysmon logs and identify process creation events (Event ID 1). Used SPL queries to filter and detect command-line activity and investigate event details, including process execution, command-line arguments, and parent-child relationships

1. Simulated endpoint activity through command-line execution
   Purpose: Demonstrates generation of system and administrative activity for detection testing

   Executed commands including:
   - whoami
   - ipconfig /all
   - netstat -ano
   - PowerShell (Get-Process, Get-Service)

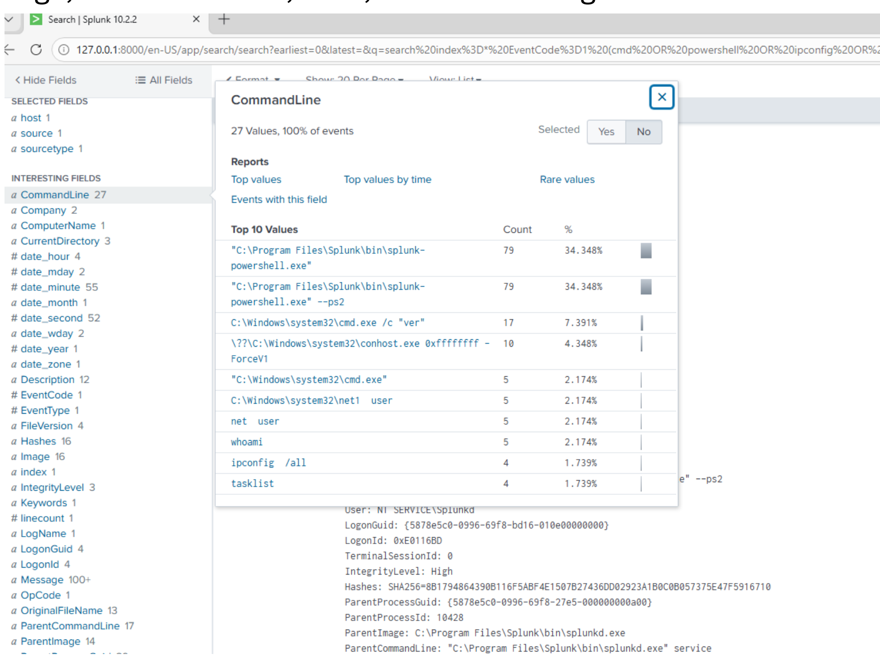


2. Detection of command-line and process execution activity using Sysmon logs in Splunk
   Purpose: Demonstrates ability to detect and analyze process creation events (Sysmon Event ID 1) using SPL queries within a SIEM platform



3. Detailed event analysis showing process execution and command-line arguments

   Purpose: Demonstrates ability to investigate security events by analyzing key fields such as Image, CommandLine, User, and ParentImage



This analysis demonstrates the ability to simulate suspicious activity and detect it using SIEM tools, replicating real-world SOC analyst workflows.